2. Homogamy: When the stamens and stigma of a flower mature at the same time it is said to be homogamy. It favours self-pollination to occur. Example: *Mirabilis jalapa, Catharanthus roseus*

3. Incomplete dichogamy: In dichogamous flowers the stamen and stigma of a flower mature at different time. Sometimes , the time of maturation of these essential organs overlap so that it becomes favourable for self-pollination.

B. Cross - pollination

It refers to the transfer of pollens on the stigma of another flower. The cross-pollination is of two types:

1. Geitonogamy: When the pollen deposits on another flower of the same individual plant, it is said to be geitonogamy. It usually occurs in plants which show monoecious condition. It is functionally cross-pollination but is generally similar to autogamy because the pollen comes from same plant.

2. Xenogamy: When the pollen (genetically different) deposits on another flower of a different plant of the same species , it is called as xenogamy.

Contrivances of cross-pollination

The flowers have several mechanisms to promote cross-pollination which are also called **contrivances of cross-pollination or outbreeding** devices. It includes the following.

1. Dicliny or Unisexuality

When the flowers are unisexual only cross-pollination is possible. There are two types.

i. **Monoecious**: Male and female flowers on the same plant. Coconut, Bitter gourd. In plants like castor and maize, autogamy is prevented but geitonogamy takes place.

ii. **Dioecious** : Male and female flowers on different plants. *Borassus*, *Carica* and Phoenix. Here both autogamy and geitonogamy are prevented.

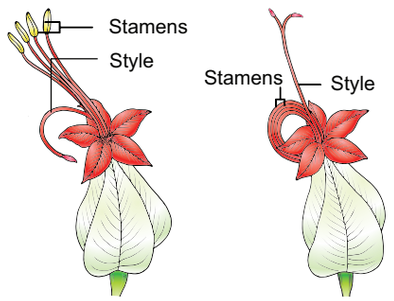
2. Monocliny or Bisexuality

Flowers are bisexual and the special adaptation of the flowers prevents self-pollination.

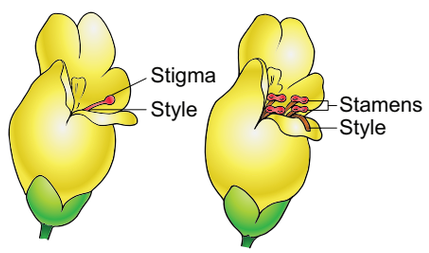
i. Dichogamy: In bisexual flowers anthers and stigmas mature at different times, thus checking self-pollination. It is of two types.

a. Protandry: The stamens mature earlier than the stigmas of the flowers. Examples: *Helianthus, Clerodendrum* (Figure 1.12 a).

b. Protogyny: The stigmas mature earlier than the stamens of the flower. Examples: *Scrophularia nodosa* and *Aristolochia bracteata* (Figure 1.12 b).



(a) Protandry - *Clerodendrum*



(b) Protogyny - *Scrophularia*

Figure 1.12 Dichogamy

ii. Herkogamy: In bisexual flowers the essential organs, the stamens and stigmas, are arranged in such a way that self-pollination becomes impossible. For example in *Gloriosa superba,* the style is reflexed away from the stamens and in *Hibiscus* the stigmas project far above the stamens (Figure 1.13).

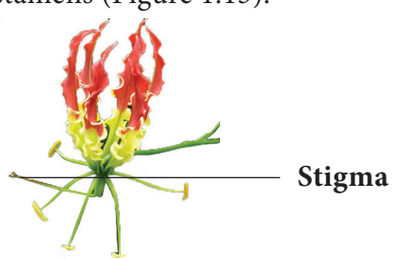


Figure 1.13 Herkogamy - *Gloriosa*

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